Experiment 1

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Subject Name: Full Stack-II **Subject Code:** 23CSH-309

1. **Aim:** To design and implement the foundational frontend architecture of the EcoTrack application using modern React practices, Vite tooling, and ES6+ JavaScript features.

2. **Objective:-**
   - To understand about basic of React and Vite.
   - To create a project using Vite with proper flow.
   - To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
   - To separate concerns using components, pages, and data modules

3. **Implementation/Code:**

   - **Logs.js :-**

```
export const logs = [
    { id: 1, activity: "Car Travel", carbon: 8 },
    { id: 2, activity: "Electricity Usage", carbon: 6 },
    { id: 3, activity: "Cycling", carbon: 0 },
    {id:  4, activity: "Bus",carbon : 3},
    {id:  5, activity: "Walking",carbon:0}
  ];
```

- **Dashboard.jsx :-**

```jsx
import { logs } from "../data/log";

const Dashboard = () => {
    const totalCarbon = logs.reduce((sum, log) => {
        return sum + log.carbon;
    }, 0);

    return (
        <div>
            <h2>Dashboard</h2>
            <p>Total Carbon Footprint: {totalCarbon} kgs</p>

            <ul>
                {logs.map((log) => (
                    <li key={log.id}>
                        {log.activity} = {log.carbon} Kg
                    </li>
                ))}
            </ul>
        </div>
    );
};
export default Dashboard;
```

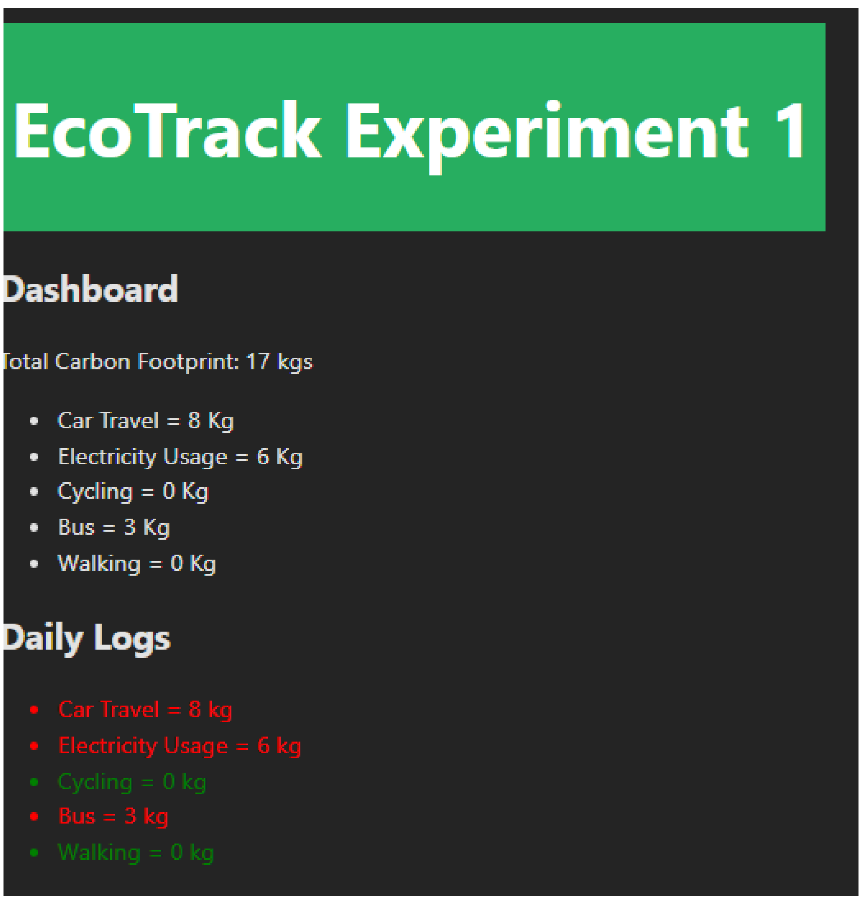
- **Logs.jsx :-**

```jsx
import React from 'react'
import { logs } from '../data/log'

export const Logs = () => {

  const highCarbon = logs.filter(
    log => log.carbon >= 4
  )

  return (
    <div>
      <h2>High Carbon Activities more than 4</h2>

      {highCarbon.map((log, index) => (
        <p key={index}>
          {log.activity} - {log.carbon} kg
        </p>
      ))}
    </div>
  )
}

export const LowCarbon = () => {
  const LowCarbon = logs.filter(
    logs =>logs.carbon <=3
  )
  return (
    <div>
      <h2>Low Carbon Activities less than 3</h2>

      {logs.map((log,index)=>{
        return (
          <p>
            {log.activity} - {log.carbon} kg
          </p>
        )
      })}
    </div>
  )
}
```

- **App.jsx:-**

```jsx
import Header from "./components/header";
import Dashboard from "./pages/dashboard";
import Logs from "./pages/logs";
const App = () => {
    return (
        <>
            <Header title="EcoTrack Experiment 1" />
            <Dashboard />
            <Logs/>
        </>
    );
};

export default App;
```

4. Output



5. Learning Outcome :-

- Created an Eco Tracker application using React to track carbon emissions.
- Displayed emission data through a dashboard built with reusable components.
- Applied JavaScript array functions to separate low and high emission activities.
- Organized the project using a component-based structure.
- Implemented dynamic rendering based on carbon emission values.
- Gained hands-on experience with data processing in React.